

A new proof of maximal theorem on Heisenberg groups

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Abstract

Given $0 \leq \alpha < 1$, we define

$$\mathbf{M}_\alpha f(u, v, t) = \sup_{\mathbf{R} \ni (0,0,0)} \text{vol}\{\mathbf{R}\}^{\alpha-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau$$

where $\mathbf{R} \subset \mathbb{R}^{2n+1}$ is a rectangle parallel to the coordinates. Moreover, $\odot$ denotes the multiplication law on a real Heisenberg group.

The $\mathbf{L}^p$ -boundedness of $\mathbf{M}_0$ has been previously proved by M. Christ. We show $\mathbf{M}_\alpha: \mathbf{L}^p(\mathbb{R}^{2n+1}) \rightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ for $\alpha = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$ by applying a geometric covering lemma due to Córdoba and Fefferman.

Consider $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$: I is an interval. $\mathbf{Q}, \mathbf{P}$ are cubes. Then, the regarding $\mathbf{L}^p \rightarrow \mathbf{L}^q$ -norm inequality is independent of n . This is a consequence of Bourgain's dimension-free estimate on Hardy-Littlewood maximal function.

1 Introduction

In this paper, $\odot$ denotes the multiplication law:

$$(u, v, t) \odot (\xi, \eta, \tau) = \left[u + \xi, v + \eta, t + \tau + \mu(u \cdot \eta - v \cdot \xi) \right], \quad \mu \in \mathbb{R} \quad (1.1)$$

on a real variable representation of Heisenberg groups for every $(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ and $(\xi, \eta, \tau)^{-1} = (-\xi, -\eta, -\tau) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$.

Let $\mathbf{R} \subset \mathbb{R}^{2n+1}$ be a rectangle parallel to the coordinates. Given $0 \leq \alpha < 1$, we define

$$\mathbf{M}_\alpha f(u, v, t) = \sup_{\mathbf{R} \ni (0,0,0)} \text{vol}\{\mathbf{R}\}^{\alpha-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau. \quad (1.2)$$

The $\mathbf{L}^p$ -boundedness of $\mathbf{M}_0$ has been proved by M. Christ [5] using a mixture of techniques developed previously by Ricci and Stein [7]-[8] and Christ [4] for singular integrals defined on sub-manifolds within a general setting of Nilpotent Lie groups.

◇ Throughout, $\mathfrak{B} > 0$ is regarded as a generic constant depending on its sub-indices.

Theorem One Let $\mathbf{M}_\alpha$ defined in (1.2) for $0 \leq \alpha < 1$. We have

$$\|\mathbf{M}_\alpha f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p n \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad \alpha = \frac{1}{p} - \frac{1}{q}, \quad 1 < p \leq q < \infty. \quad (1.3)$$

We prove **Theorem One** by applying a geometric covering lemma due to Córdoba and Fefferman [3].

Córdoba-Fefferman covering lemma Let $\mathbf{R}_j, j = 1, 2, \dots$ be a sequence of rectangles in $\mathbb{R}^{2n+1}$ parallel to the coordinates. There exists a subsequence $\widehat{\mathbf{R}}_k, k = 1, 2, \dots$ such that

$$\text{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \leq \mathfrak{B}_n \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad (1.4)$$

and

$$\left\|\sum_k \chi_{\widehat{\mathbf{R}}_k}\right\|_{L^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}, \quad 1 < p < \infty \quad (1.5)$$

where χ is an indicator function.

Remark 1.1. This covering lemma is true for any given absolutely continuous measure satisfying the $\mathbf{A}_\infty$ -property uniformly on every coordinate subspace. Its proof relies on a beautiful 'slicing' idea which reduces our assertion to a lower dimensional subspace.

For the Lebesgue measure case, we can prove this result with a more direct approach.

Next, consider $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ of which I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes. We show the above covering lemma hold independently from n by applying Bourgain's dimension-free estimate on Hardy-Littlewood maximal function [2].

Dimension-free covering lemma Let $\mathbf{R}_j = \mathbf{Q}_j \times \mathbf{P}_j \times I_j \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ for every $j = 1, 2, \dots$. I_j is an interval and $\mathbf{Q}_j, \mathbf{P}_j$ are cubes. There exists a subsequence $\widehat{\mathbf{R}}_k, k = 1, 2, \dots$ such that

$$\text{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \leq \mathfrak{B} \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad (1.6)$$

and

$$\left\|\sum_k \chi_{\widehat{\mathbf{R}}_k}\right\|_{L^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}, \quad 1 < p < \infty. \quad (1.7)$$

By using the above covering lemma instead, we obtain our second result.

Theorem Two Let $\mathbf{M}_\alpha$ defined in (1.2) for $0 \leq \alpha < 1$ over rectangles $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$. We have

$$\|\mathbf{M}_\alpha f\|_{L^q(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \|f\|_{L^p(\mathbb{R}^{2n+1})}, \quad \alpha = \frac{1}{p} - \frac{1}{q}, \quad 1 < p \leq q < \infty. \quad (1.8)$$

Certain dimension-free estimates trace back to the earlier work of Stein and Strömberg [6] for Hardy-Littlewood maximal function defined over Euclidean balls. Bourgain [1] gives a generalization by replacing the balls with symmetric convex bodies restricted to $p > \frac{3}{2}$. Later, Müller [9] improves this result to every $1 < p < \infty$. The extreme case for which the maximal operator is defined over cubes is eventually proved by Bourgain [2]. Recently, such a dimension-free estimate is established by Ganguly and Ghosh [10] for vector-value Hardy-Littlewood maximal function on Heisenberg groups.

The remaining paper is organized as follows. In the next section, we prove **Theorem One** and **Theorem Two**. In section 3, we give a simpler proof of **Córdoba-Fefferman covering lemma** and its dimension-free version.

2 Proof of Theorem One and Theorem Two

Recall $\mathbf{M}_\alpha$ defined in (1. 2) for $0 \leq \alpha < 1$. By taking $\xi \longrightarrow u - \xi$, $\eta \longrightarrow v - \eta$ and $\tau \longrightarrow t - \tau$, $\mathbf{M}_\alpha$ can be equivalently defined as

$$\mathbf{M}_\alpha f(u, v, t) = \sup_{\mathbf{R} \ni (u, v, t)} \text{vol}\{\mathbf{R}\}^{\alpha-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau. \quad (2. 1)$$

Let $\alpha = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$. Consider

$$\mathbf{U}_\lambda = \left\{ (u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\alpha f(u, v, t) > \lambda \right\}. \quad (2. 2)$$

Given any $(u, v, t) \in \mathbf{U}_\lambda$, there is a rectangle $\mathbf{R}_j \ni (u, v, t)$ such that

$$\text{vol}\{\mathbf{R}_j\}^{\alpha-1} \iiint_{\mathbf{R}_j} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau > \frac{1}{2} \lambda. \quad (2. 3)$$

Let (u, v, t) run through the set $\mathbf{U}_\lambda$. We have $\mathbf{U}_\lambda \subset \bigcup_j \mathbf{R}_j$. By applying **Córdoba-Fefferman covering lemma**, we can select a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}^\infty$ from the union above and

$$\text{vol}\{\mathbf{U}_\lambda\} \lesssim \text{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \leq \mathfrak{B}_n \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad \text{by (1. 4)}. \quad (2. 4)$$

Because $0 \leq \alpha < 1$, we further have

$$\begin{aligned} \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} &\leq \sum_k \text{vol}\{\widehat{\mathbf{R}}_k\} \\ &\lesssim \lambda^{-\frac{1}{1-\alpha}} \left\{ \sum_k \iiint_{\widehat{\mathbf{R}}_k} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \right\}^{\frac{1}{1-\alpha}} \quad \text{by (2. 3)} \\ &= \lambda^{-\frac{1}{1-\alpha}} \left\{ \iiint_{\mathbb{R}^{2n+1}} \left| f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi)) \sum_k \chi_{\widehat{\mathbf{R}}_k}(\xi, \eta, \tau) \right| d\xi d\eta d\tau \right\}^{\frac{1}{1-\alpha}} \\ &\leq \lambda^{-\frac{1}{1-\alpha}} \left\{ \iiint_{\mathbb{R}^{2n+1}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))|^p d\xi d\eta d\tau \right\}^{\frac{1}{p} \frac{1}{1-\alpha}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\alpha}} \\ &\quad \text{by Hölder inequality} \\ &= \lambda^{-\frac{1}{1-\alpha}} \left\{ \iint_{\mathbb{R}^{2n}} \|f(\xi, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})}^p d\xi d\eta \right\}^{\frac{1}{p} \frac{1}{1-\alpha}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\alpha}} \\ &\leq (\mathfrak{B}_p n)^{\frac{1}{1-\alpha}} \lambda^{-\frac{1}{1-\alpha}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}^{\frac{1}{1-\alpha}} \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{p-1}{p} \frac{1}{1-\alpha}} \quad \text{by (1. 5)}. \end{aligned} \quad (2. 5)$$

By raising both sides of (2. 5) to the $(1 - \alpha)$ -th power and then taking into account for $1 - \alpha - \frac{p-1}{p} = \frac{1}{p} - \left[\frac{1}{p} - \frac{1}{q} \right] = \frac{1}{q}$, we find

$$\text{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{q}} \leq \mathfrak{B}_p n \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \quad (2. 6)$$

Let $\mathbf{U}_\lambda$ defined in (2. 2). From (2. 4) and (2. 6), we obtain

$$\begin{aligned} \text{vol} \left\{ (u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\alpha f(u, v, t) > \lambda \right\}^{\frac{1}{q}} &\leq \text{vol} \left\{ \bigcup_j \mathbf{R}_j \right\}^{\frac{1}{q}} \\ &\leq \mathfrak{B}_n \text{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}^{\frac{1}{q}} \\ &\leq \mathfrak{B}_p n \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \end{aligned} \quad (2. 7)$$

By using this weak type (p, q) -estimate and applying Marcinkiewicz interpolation theorem, we conclude (1. 3).

Now, assert $\mathbf{M}_\alpha$ defined over rectangles $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$ of which I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes. By carrying out (2. 4)-(2. 6) as above and applying (1. 6)-(1. 7) from **Dimension-free covering lemma** instead of (1. 4)-(1. 5), we obtain

$$\text{vol} \left\{ (u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\alpha f(u, v, t) > \lambda \right\}^{\frac{1}{q}} \leq \mathfrak{B}_p \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})} \quad (2. 8)$$

where the implied constant is independent of n .

By using the dimension-free weak type (p, q) -estimate in (2. 8) and applying Marcinkiewicz interpolation theorem, we conclude (1. 8).

3 Proof of the covering lemmas

Given $\mathbf{R}_j, j = 1, 2, \dots$ which are rectangles in $\mathbb{R}^{2n+1}$ parallel to the coordinates, we select $\widehat{\mathbf{R}}_k, k = 1, 2, \dots$ as follows.

Let $\widehat{\mathbf{R}}_1 = \mathbf{R}_1$. Having chosen $\widehat{\mathbf{R}}_1, \widehat{\mathbf{R}}_2, \dots, \widehat{\mathbf{R}}_{k-1}$, we pick $\widehat{\mathbf{R}}_k$ as the first rectangle $\mathbf{R}$ on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_{k-1}$ so that

$$\text{vol} \left\{ \mathbf{R} \cap \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell \right\} \leq \frac{1}{2} \text{vol} \{ \mathbf{R} \}. \quad (3. 1)$$

Suppose that $\mathbf{R}$ is an unselected rectangle. There is a positive number N such that $\mathbf{R}$ is on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_N$ and

$$\text{vol} \left\{ \mathbf{R} \cap \bigcup_{k=1}^N \widehat{\mathbf{R}}_k \right\} > \frac{1}{2} \text{vol} \{ \mathbf{R} \}. \quad (3. 2)$$

Let $\mathbf{M}$ be the strong maximal operator defined as

$$\mathbf{M}f(u, v, t) = \sup_{\mathbf{R} \ni (u, v, t)} \text{vol}\{\mathbf{R}\}^{-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau)| d\xi d\eta d\tau. \quad (3.3)$$

Here, $\mathbf{R}$ is an $2n + 1$ -parameter rectangle. A simple observation is

$$\mathbf{M} \leq \mathbf{M}^{(1)} \dots \mathbf{M}^{(1)} \quad (3.4)$$

where the right-hand-side of (3.4) is an $(2n + 1)$ -fold composition of Hardy-Littlewood maximal operators: $\mathbf{M}^{(1)}$ defined in every coordinate subspace. A $(2n + 1)$ -fold iteration estimate gives

$$\|\mathbf{M}f\|_{L^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \, n \, \|f\|_{L^p(\mathbb{R}^{2n+1})}, \quad 1 < p < \infty. \quad (3.5)$$

From (3.2), we find

$$\mathbf{M}\chi_{\bigcup_k \widehat{\mathbf{R}}_k}(u, v, t) > \frac{1}{2}, \quad (u, v, t) \in \bigcup_j \mathbf{R}_j. \quad (3.6)$$

By applying the L^p -estimate in (3.5), we have

$$\begin{aligned} \text{vol}\left\{\bigcup_j \mathbf{R}_j\right\} &= \iiint_{\bigcup_j \mathbf{R}_j} dudvdt \\ &\leq 2^2 \iiint_{\bigcup_j \mathbf{R}_j} [\mathbf{M}\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \quad \text{by (3.6)} \\ &\lesssim \iiint_{\mathbb{R}^{2n+1}} [\mathbf{M}\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &\leq \mathfrak{B}_n \iiint_{\mathbb{R}^{2n+1}} [\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &= \mathfrak{B}_n \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} dudvdt \\ &= \mathfrak{B}_n \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}. \end{aligned} \quad (3.7)$$

On the other hand, (3.1) suggests

$$\text{vol}\left\{\widehat{\mathbf{R}}_k \cap \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell\right\} \leq \frac{1}{2} \text{vol}\{\widehat{\mathbf{R}}_k\}, \quad k = 1, 2, \dots \quad (3.8)$$

which further implies

$$\text{vol}\left\{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell\right\} > \frac{1}{2} \text{vol}\{\widehat{\mathbf{R}}_k\}, \quad k = 1, 2, \dots \quad (3.9)$$

Let $\phi \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})$ and $\|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} = 1$. We have

$$\begin{aligned}
& \iiint_{\mathbb{R}^{2n+1}} \phi(\xi, \eta, \tau) \sum_k \chi_{\widehat{\mathbf{R}}_k}(\xi, \eta, \tau) d\xi d\eta d\tau \\
&= \sum_k \iiint_{\widehat{\mathbf{R}}_k} \phi(\xi, \eta, \tau) d\xi d\eta d\tau \\
&= \sum_k \left\{ \mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1} \iiint_{\widehat{\mathbf{R}}_k} \phi(\xi, \eta, \tau) d\xi d\eta d\tau \right\} \mathbf{vol}\{\widehat{\mathbf{R}}_k\} \\
&< 2 \sum_k \left\{ \mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1} \iiint_{\widehat{\mathbf{R}}_k} |\phi(\xi, \eta, \tau)| d\xi d\eta d\tau \right\} \mathbf{vol}\left\{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell\right\} \quad \text{by (3. 9)} \quad (3. 10) \\
&\lesssim \sum_k \int_{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell} \left\{ \mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1} \iiint_{\widehat{\mathbf{R}}_k} |\phi(\xi, \eta, \tau)| d\xi d\eta d\tau \right\} dudvdt \\
&\leq \sum_k \iiint_{\widehat{\mathbf{R}}_k \setminus \bigcup_{\ell=1}^{k-1} \widehat{\mathbf{R}}_\ell} \mathbf{M}\phi(u, v, t) dudvdt \\
&= \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}\phi(u, v, t) dudvdt.
\end{aligned}$$

Hölder inequality further implies

$$\begin{aligned}
\iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}\phi(u, v, t) dudvdt &\leq \|\mathbf{M}\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}} \\
&\leq \mathfrak{B}_{p \ n} \|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}} \quad \text{by (3. 5)} \quad (3. 11) \\
&= \mathfrak{B}_{p \ n} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}.
\end{aligned}$$

By substituting (3. 11) to (3. 10) and taking the supremum of ϕ , we arrive at

$$\left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_{p \ n} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}, \quad 1 < p < \infty. \quad (3. 12)$$

Now, we consider $\mathbf{M}$ defined in (3. 3) over rectangles $\mathbf{R} = \mathbf{Q} \times \mathbf{P} \times I \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$. Note that I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes. In this case, we find

$$\mathbf{M} \leq \mathbf{M}^{(n)} \mathbf{M}^{(n)} \mathbf{M}^{(1)} \quad (3. 13)$$

where the right-hand-side of (3. 13) is a 3-fold composition of Hardy-Littlewood maximal operator defined respectively in $\mathbb{R}^n$ and $\mathbb{R}$.

Recall the elegant work of Bourgain [2]: We have

$$\|\mathbf{M}^{(n)}f\|_{L^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \|f\|_{L^p(\mathbb{R}^{2n+1})}, \quad 1 < p < \infty \quad (3.14)$$

where the implied constant is independent of n .

By using (3.14) and carrying out a 3-fold iteration, we conclude that $\mathbf{M}$ satisfies the same dimension-free L^p -estimate. A repeat argument of (3.6)-(3.12) concludes the same results in (3.7) and (3.12) except that the implied constants $\mathfrak{B}_n, \mathfrak{B}_{p,n}$ will be replaced by $\mathfrak{B}, \mathfrak{B}_p$.

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